TriNetra-AMRF — Phase 6: Tracking Validation Results

Protocol. Four layers of evidence, each labelled for what it can and cannot show: (1) unit-tested metric and tracker implementations; (2) a synthetic benchmark with exact ground truth, whose detector simulation couples quality to TRC (with controls where the gate has no reason to help); (3) label-free quality metrics on the 14 real RGB-thermal clips (no track annotations exist); (4) a thermal-degradation stress test on real footage, measuring both the TRC–detector coupling and identity stability. No ground-truth accuracy (MOTA/IDF1/HOTA against human labels) is reported for the real clips.

Findings:

- Severity 0.5: TRC gate vs ByteTrack on HOTA: no significant difference (Δ -0.000, 95% CI [-0.005, +0.005]).
- Severity 0.5: TRC gate vs ByteTrack on IDF1: no significant difference (Δ -0.002, 95% CI [-0.011, +0.006]).
- Severity 0.5: TRC gate vs ByteTrack on MOTA: significantly worse (Δ -0.003, 95% CI [-0.004, -0.002]).
- Severity 0.5: TRC gate vs ByteTrack on IDSW: no significant difference (Δ +0.167, 95% CI [-0.433, +0.733]).
- Control 'uninformative' (severity 0.5): gate vs ByteTrack HOTA significantly worse (Δ -0.001).
- Control 'no_degradation' (severity 0.5): gate vs ByteTrack HOTA significantly worse (Δ -0.000).
- Severity 1.0: TRC gate vs ByteTrack on HOTA: significantly worse (Δ -0.003, 95% CI [-0.005, -0.000]).
- Severity 1.0: TRC gate vs ByteTrack on IDF1: no significant difference (Δ -0.004, 95% CI [-0.009, +0.000]).
- Severity 1.0: TRC gate vs ByteTrack on MOTA: significantly worse (Δ -0.003, 95% CI [-0.005, -0.002]).
- Severity 1.0: TRC gate vs ByteTrack on IDSW: no significant difference (Δ +0.033, 95% CI [-0.433, +0.467]).
- Control 'uninformative' (severity 1.0): gate vs ByteTrack HOTA significantly worse (Δ -0.001).
- Control 'no_degradation' (severity 1.0): gate vs ByteTrack HOTA no significant difference (Δ -0.001).
- Real clips (8): the TriNetra tracker produced 296 tracks (median length 61 frames, coverage 0.97); ByteTrack without the gate produced 299.
- Real stress test: gate vs ByteTrack identity stability on IDF1: no significant difference (Δ +0.003, n = 6).
- Real stress test: gate vs ByteTrack identity stability on HOTA: no significant difference (Δ +0.007, n = 6).
- Real stress test: gate vs ByteTrack identity stability on AssA: no significant difference (Δ +0.013, n = 6).

Implementation validation

Suite	Checks passed
tracker / gate / behaviour self-test	28/28
metric implementation self-test	17/17

Metric tests compare against values derived by hand (e.g. a two-track ID swap gives IDF1 = 0.5, HOTA = 0.577); tracker tests cover lifecycle, ID stability, low-confidence recovery, the TRC gate and behaviour flags.

Synthetic benchmark, severity 0.5 — TRC-coupled degradation (30 sequences, exact ground truth)

Tracker	HOTA	DetA	AssA	MOTA	IDF1	IDSW	FP	FN
SORT-style	0.556 ± 0.056	0.602 ± 0.036	0.515 ± 0.079	0.745 ± 0.041	0.752 ± 0.075	12.8 ± 9.5	17.2 ± 10.1	769.2 ± 202.6
DeepSORT (motion)	0.514 ± 0.052	0.576 ± 0.032	0.461 ± 0.075	0.702 ± 0.042	0.697 ± 0.072	15.9 ± 10.4	155.3 ± 53.0	755.4 ± 184.9
ByteTrack	0.559 ± 0.057	0.608 ± 0.034	0.516 ± 0.084	0.754 ± 0.038	0.754 ± 0.078	12.4 ± 9.3	17.6 ± 10.5	739.3 ± 190.6
ByteTrack + TRC gate (ours)	0.559 ± 0.056	0.607 ± 0.035	0.516 ± 0.080	0.751 ± 0.039	0.752 ± 0.076	12.5 ± 9.4	17.0 ± 10.1	748.9 ± 195.5

mean ± std over seeded sequences; identical detections and TRC for every tracker.

Paired comparison, ours minus baseline (severity 0.5, coupled)

Baseline	Metric	Mean Δ	95% CI (bootstrap)	Wins/Losses	Wilcoxon p	Reading
ByteTrack	HOTA	-0.000	[-0.005, +0.005]	4/25	0.00446	no significant difference
ByteTrack	IDF1	-0.002	[-0.011, +0.006]	5/24	0.00583	no significant difference
ByteTrack	MOTA	-0.003	[-0.004, -0.002]	1/28	5.9e-06	significantly worse
ByteTrack	IDSW	+0.167	[-0.433, +0.733]	5/3	0.574	no significant difference
SORT-style	HOTA	+0.003	[-0.000, +0.004]	29/1	3.79e-06	no significant difference
SORT-style	IDF1	+0.000	[-0.005, +0.003]	29/1	3.79e-06	no significant difference
SORT-style	MOTA	+0.007	[+0.006, +0.008]	30/0	1.86e-09	significantly better
SORT-style	IDSW	-0.233	[-0.600, +0.100]	2/5	0.23	no significant difference
DeepSORT (motion)	HOTA	+0.044	[+0.029, +0.059]	26/4	1.82e-05	significantly better
DeepSORT (motion)	IDF1	+0.055	[+0.030, +0.078]	25/5	0.000153	significantly better
DeepSORT (motion)	MOTA	+0.049	[+0.042, +0.056]	30/0	1.86e-09	significantly better
DeepSORT (motion)	IDSW	-3.400	[-7.267, +0.767]	7/23	0.0248	no significant difference

Δ = ours − baseline over the same sequences; for IDSW lower is better.

Control: uninformative (severity 0.5)

Tracker	HOTA	IDF1	MOTA	IDSW
SORT-style	0.553 ± 0.059	0.743 ± 0.081	0.747 ± 0.041	15.4 ± 8.4
DeepSORT (motion)	0.526 ± 0.050	0.713 ± 0.080	0.709 ± 0.044	17.0 ± 9.2
ByteTrack	0.560 ± 0.055	0.751 ± 0.075	0.757 ± 0.039	15.3 ± 8.2
ByteTrack + TRC gate (ours)	0.559 ± 0.055	0.750 ± 0.075	0.755 ± 0.039	15.3 ± 8.3

uninformative: detector degrades but the tracker receives a constant TRC. no_degradation: TRC dips but the detector is unaffected.

Control: no_degradation (severity 0.5)

Tracker	HOTA	IDF1	MOTA	IDSW
SORT-style	0.650 ± 0.046	0.833 ± 0.067	0.826 ± 0.026	6.9 ± 5.0
DeepSORT (motion)	0.601 ± 0.048	0.774 ± 0.066	0.789 ± 0.031	11.8 ± 12.8
ByteTrack	0.652 ± 0.045	0.834 ± 0.067	0.828 ± 0.026	6.9 ± 5.0
ByteTrack + TRC gate (ours)	0.651 ± 0.045	0.834 ± 0.067	0.828 ± 0.026	6.9 ± 5.0

uninformative: detector degrades but the tracker receives a constant TRC. no_degradation: TRC dips but the detector is unaffected.

Gate strength β sweep (severity 0.5, coupled)

β	HOTA	IDF1	MOTA	IDSW	FN
0.0	0.559 ± 0.057	0.754 ± 0.078	0.754 ± 0.038	12.4	739
0.25	0.559 ± 0.059	0.753 ± 0.080	0.753 ± 0.039	12.6	744
0.5	0.559 ± 0.056	0.752 ± 0.076	0.751 ± 0.039	12.5	749
1.0	0.559 ± 0.056	0.754 ± 0.076	0.749 ± 0.040	12.7	755
2.0	0.558 ± 0.056	0.753 ± 0.076	0.748 ± 0.041	12.7	760
4.0	0.557 ± 0.056	0.752 ± 0.076	0.746 ± 0.041	12.7	765

β = 0 disables the gate (plain ByteTrack).

Degradation-severity sweep (coupled; HOTA by tracker)

Severity	Frames with TRC<0.6	SORT-style	DeepSORT (motion)	ByteTrack	ByteTrack + TRC gate (ours)	IDSW ByteTrack → ours
0.0	0.30	0.758	0.703	0.758	0.758	4.3 → 4.3
0.25	0.30	0.650	0.606	0.651	0.651	9.8 → 9.5
0.5	0.30	0.556	0.514	0.559	0.559	12.4 → 12.5
0.75	0.30	0.407	0.428	0.418	0.413	20.5 → 20.8
1.0	0.30	0.319	0.359	0.326	0.323	17.6 → 17.7
1.5	0.30	0.230	0.232	0.236	0.235	21.7 → 21.8

Synthetic benchmark, severity 1.0 — TRC-coupled degradation (30 sequences, exact ground truth)

Tracker	HOTA	DetA	AssA	MOTA	IDF1	IDSW	FP	FN
SORT-style	0.319 ± 0.082	0.380 ± 0.077	0.272 ± 0.092	0.471 ± 0.101	0.467 ± 0.119	17.8 ± 11.6	25.3 ± 13.2	1611.8 ± 446.3
DeepSORT (motion)	0.359 ± 0.059	0.412 ± 0.056	0.317 ± 0.075	0.456 ± 0.095	0.525 ± 0.092	27.0 ± 18.7	357.5 ± 94.1	1309.6 ± 334.1
ByteTrack	0.326 ± 0.083	0.388 ± 0.076	0.278 ± 0.094	0.480 ± 0.101	0.476 ± 0.121	17.6 ± 11.5	32.5 ± 18.7	1576.4 ± 434.4
ByteTrack + TRC gate (ours)	0.323 ± 0.082	0.385 ± 0.077	0.275 ± 0.092	0.476 ± 0.102	0.471 ± 0.119	17.7 ± 11.5	30.1 ± 17.8	1590.0 ± 440.6

mean ± std over seeded sequences; identical detections and TRC for every tracker.

Paired comparison, ours minus baseline (severity 1.0, coupled)

Baseline	Metric	Mean Δ	95% CI (bootstrap)	Wins/Losses	Wilcoxon p	Reading
ByteTrack	HOTA	-0.003	[-0.005, -0.000]	4/26	0.00104	significantly worse
ByteTrack	IDF1	-0.004	[-0.009, +0.000]	8/22	0.0106	no significant difference
ByteTrack	MOTA	-0.003	[-0.005, -0.002]	3/27	1.42e-06	significantly worse
ByteTrack	IDSW	+0.033	[-0.433, +0.467]	9/9	0.771	no significant difference
SORT-style	HOTA	+0.004	[+0.002, +0.006]	28/2	1.6e-05	significantly better
SORT-style	IDF1	+0.005	[+0.001, +0.008]	26/4	0.000209	significantly better
SORT-style	MOTA	+0.005	[+0.003, +0.007]	27/2	0.000114	significantly better
SORT-style	IDSW	-0.167	[-0.767, +0.400]	8/8	0.548	no significant difference
DeepSORT (motion)	HOTA	-0.037	[-0.054, -0.018]	6/24	0.00113	significantly worse
DeepSORT (motion)	IDF1	-0.054	[-0.087, -0.019]	7/23	0.00501	significantly worse
DeepSORT (motion)	MOTA	+0.021	[+0.007, +0.034]	20/10	0.00664	significantly better
DeepSORT (motion)	IDSW	-9.367	[-13.467, -5.600]	4/22	0.000241	significantly better

Δ = ours − baseline over the same sequences; for IDSW lower is better.

Control: uninformative (severity 1.0)

Tracker	HOTA	IDF1	MOTA	IDSW
SORT-style	0.321 ± 0.068	0.475 ± 0.098	0.470 ± 0.088	18.8 ± 7.9
DeepSORT (motion)	0.352 ± 0.048	0.514 ± 0.067	0.452 ± 0.091	30.3 ± 16.1
ByteTrack	0.328 ± 0.068	0.481 ± 0.099	0.478 ± 0.088	19.2 ± 8.5
ByteTrack + TRC gate (ours)	0.327 ± 0.067	0.479 ± 0.098	0.477 ± 0.088	19.0 ± 8.6

uninformative: detector degrades but the tracker receives a constant TRC. no_degradation: TRC dips but the detector is unaffected.

Control: no_degradation (severity 1.0)

Tracker	HOTA	IDF1	MOTA	IDSW
SORT-style	0.548 ± 0.036	0.761 ± 0.059	0.744 ± 0.028	10.2 ± 9.0
DeepSORT (motion)	0.506 ± 0.036	0.707 ± 0.066	0.695 ± 0.030	14.4 ± 12.7
ByteTrack	0.552 ± 0.037	0.765 ± 0.060	0.752 ± 0.026	10.3 ± 10.0
ByteTrack + TRC gate (ours)	0.552 ± 0.036	0.765 ± 0.059	0.751 ± 0.026	10.2 ± 9.1

uninformative: detector degrades but the tracker receives a constant TRC. no_degradation: TRC dips but the detector is unaffected.

Gate strength β sweep (severity 1.0, coupled)

β	HOTA	IDF1	MOTA	IDSW	FN
0.0	0.326 ± 0.083	0.476 ± 0.121	0.480 ± 0.101	17.6	1576
0.25	0.324 ± 0.082	0.472 ± 0.120	0.478 ± 0.101	17.6	1585
0.5	0.323 ± 0.082	0.471 ± 0.119	0.476 ± 0.102	17.7	1590
1.0	0.321 ± 0.082	0.470 ± 0.119	0.474 ± 0.102	17.8	1596
2.0	0.321 ± 0.081	0.468 ± 0.118	0.473 ± 0.102	17.8	1600
4.0	0.320 ± 0.082	0.467 ± 0.119	0.472 ± 0.102	17.7	1605

β = 0 disables the gate (plain ByteTrack).

Real footage: per-clip tracking quality (TriNetra tracker; label-free metrics)

Clip	Frames	TRC mean	TRC min	Tracks	Median len	Short (<1 s)	Coverage	Frag. rate	Jumps/1k	Recoveries	Behaviour flags (tracks)
seq 01	1500	0.78	0.78	50	56	0.12	0.98	0.08	0.1	22	—
seq 02	1500	0.76	0.75	29	76	0.10	0.97	0.03	0.0	30	—
seq 03	1500	0.78	0.78	15	101	0.13	0.97	0.13	0.0	17	—
seq 04	1500	0.78	0.77	12	22	0.58	0.98	0.42	0.0	2	—
seq 05	1500	0.78	0.78	28	26	0.39	0.98	0.29	0.2	19	—
seq 06	1500	0.75	0.72	56	46	0.39	0.95	0.09	0.0	71	—
seq 07	1500	0.72	0.68	66	64	0.21	0.96	0.30	0.3	94	—
seq 08	1500	0.75	0.73	40	95	0.15	0.98	0.07	0.0	40	—

Label-free proxies (no ground truth exists for these clips). Coverage = share of confident detections explained by a track; Frag. rate = suspected identity fragmentations per track.

Real footage: tracker comparison across all clips (mean ± std over clips)

Tracker	Total tracks	Median len	Short ratio	Coverage	Frag. rate	Jumps/1k	Recoveries	ms/frame
SORT-style	319	44.5 ± 19.5	0.40 ± 0.10	0.97 ± 0.01	0.19 ± 0.09	0.2 ± 0.2	1717	0.38
DeepSORT (motion)	509	28.0 ± 11.3	0.45 ± 0.13	0.98 ± 0.01	0.32 ± 0.10	0.1 ± 0.2	n/a	n/a
ByteTrack	299	62.1 ± 29.7	0.27 ± 0.18	0.97 ± 0.01	0.19 ± 0.12	0.1 ± 0.1	232	0.43
ByteTrack + TRC gate (ours)	296	60.9 ± 29.2	0.26 ± 0.18	0.97 ± 0.01	0.18 ± 0.14	0.1 ± 0.1	295	0.43

Same cached detections for every tracker. Fewer tracks / shorter-track ratio / fragmentation at equal coverage indicate more consolidated identities; none of these prove correctness.

Real-footage coupling: detector behaviour vs frame TRC (thermal damaged, RGB intact)

TRC bin	Frames	Recall vs clean	Mean IoU (matched)	Score ratio	Detections/frame ratio
0.40-0.60	300	0.89	0.89	0.80	0.74
0.60-0.75	1500	0.97	0.90	0.93	0.88

Reference = confident detections (score ≥ 0.6) of the clean run. This measures whether low TRC actually goes with worse detections — the assumption behind the gate.

Identity stability under thermal degradation (6 runs, degraded segment only)

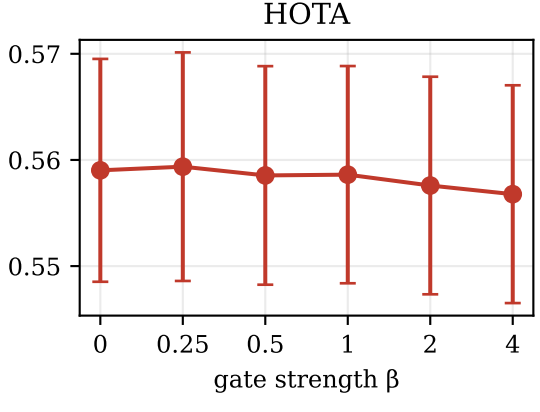
Tracker	IDF1	HOTA	AssA	IDSW
SORT-style	0.733 ± 0.124	0.630 ± 0.122	0.654 ± 0.178	5.2 ± 2.4
DeepSORT (motion)	0.658 ± 0.083	0.592 ± 0.071	0.538 ± 0.120	38.0 ± 22.7
ByteTrack	0.790 ± 0.085	0.695 ± 0.090	0.696 ± 0.151	6.2 ± 4.1
ByteTrack + TRC gate (ours)	0.793 ± 0.082	0.702 ± 0.071	0.709 ± 0.131	6.2 ± 3.9

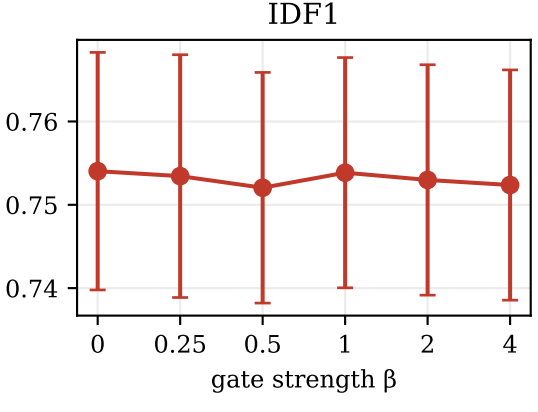
Each tracker's degraded-run tracks are scored against its own clean-run tracks (self-consistency; the reference is a tracker output, not human annotation).

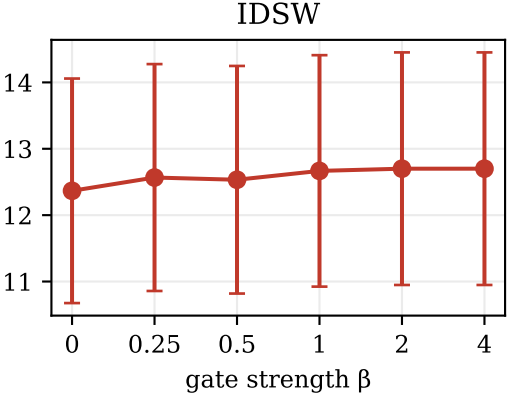
Stress test: ours minus ByteTrack (paired over runs)

Metric	Mean Δ	95% CI (bootstrap)	Wins/Losses	Reading
IDF1	+0.003	[-0.041, +0.045]	3/3	no significant difference
HOTA	+0.007	[-0.020, +0.042]	3/3	no significant difference
AssA	+0.013	[-0.033, +0.070]	2/4	no significant difference
IDSW	+0.000	[-1.000, +1.000]	2/2	no significant difference

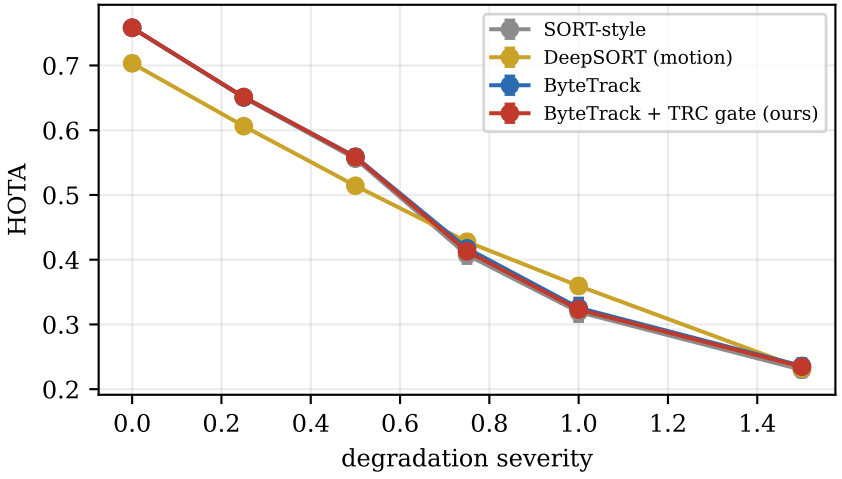
β sensitivity (severity 0.5, error bars = s.e.m.)

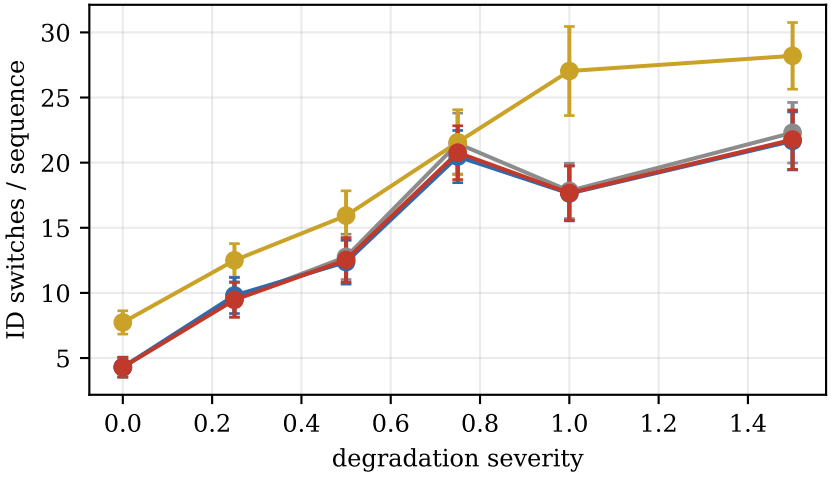


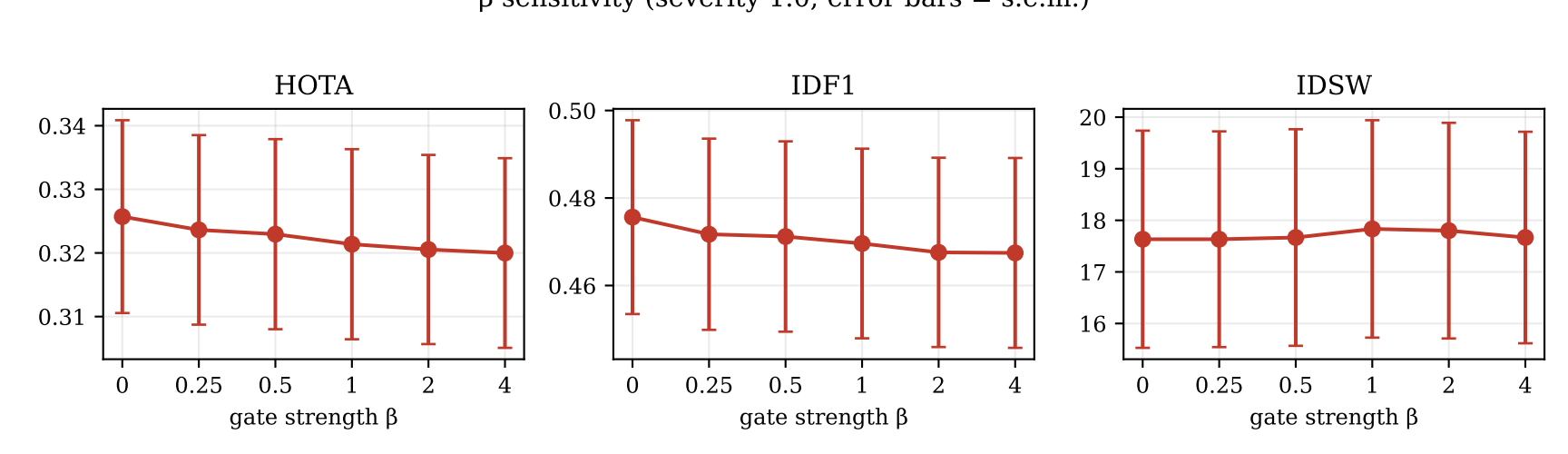


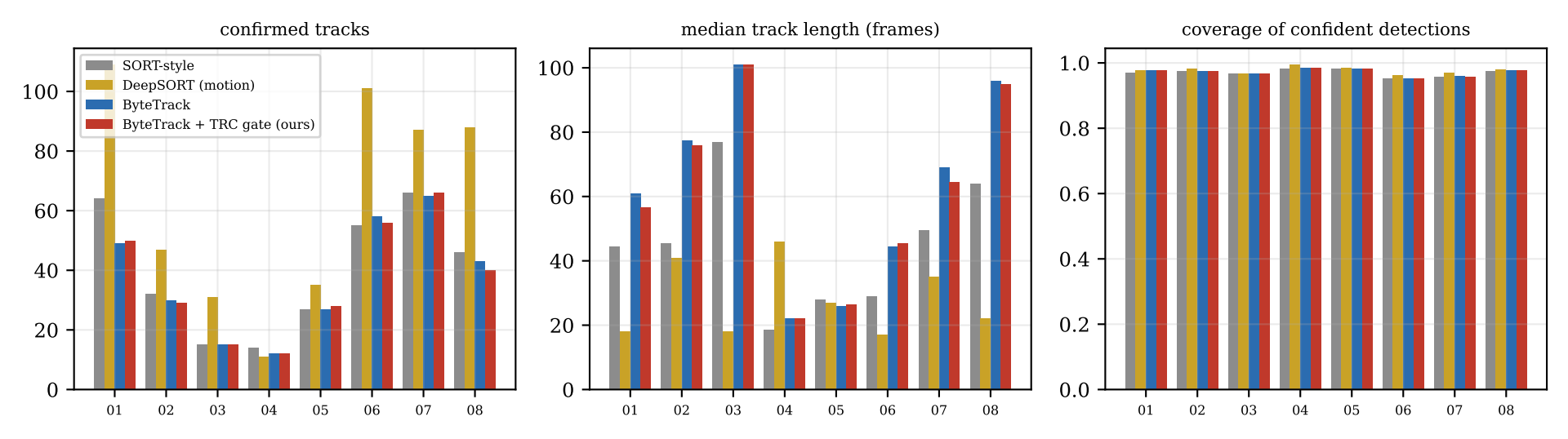


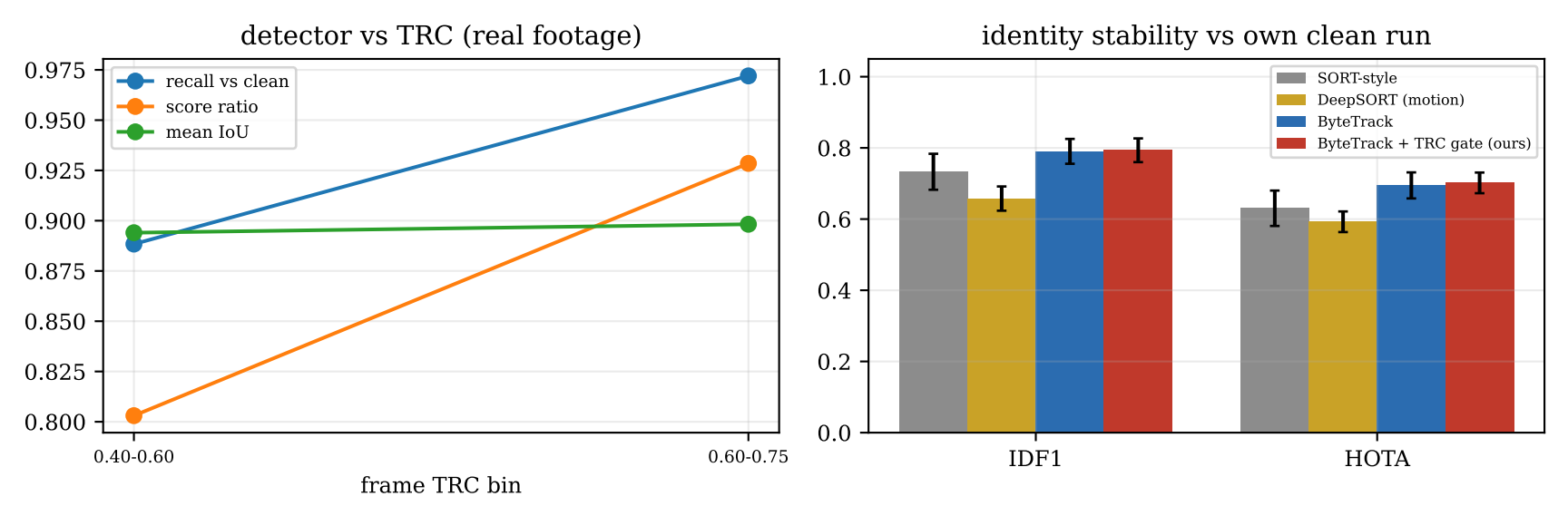
Synthetic benchmark: robustness to detector degradation (error bars = s.e.m.)






β sensitivity (severity 1.0, error bars = s.e.m.)
HOTA
0.34
0.33
0.32
0.31
0
0.25
0.5
1
2
4
gate strength β
IDF1
0.50
0.48
0.46
0
0.25
0.5
1
2
4
gate strength β
IDSW
20
19
18
17
16
0
0.25
0.5
1
2
4
gate strength β


confirmed tracks
SORT-style
DeepSORT (motion)
ByteTrack
ByteTrack + TRC gate (ours)
0
20
40
60
80
100
01
02
03
04
05
06
07
08
median track length (frames)
0
20
40
60
80
100
01
02
03
04
05
06
07
08
coverage of confident detections
0.0
0.2
0.4
0.6
0.8
1.0
01
02
03
04
05
06
07
08


detector vs TRC (real footage)
recall vs clean
score ratio
mean IoU
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0.800
0.40-0.60
0.60-0.75
frame TRC bin
identity stability vs own clean run
SORT-style
DeepSORT (motion)
ByteTrack
ByteTrack + TRC gate (ours)
1.0
0.8
0.6
0.4
0.2
0.0
IDF1
HOTA